

Fall 2002

Problem 1 (Fa02). Let G be a finite non-abelian group of order n . Show that there exists an integer d satisfying $2 \leq d \leq n/2$ and a set P of cardinality d , such that G acts transitively on P .

Problem 2 (Fa02). Let $\{x_{ij} \mid i, j \in \mathbb{N}\}$ be a doubly indexed set in a complete metric space (X, ρ) such that

$$\rho(x_{ij}, x_{k\ell}) \leq \min \left\{ \max \left\{ \frac{1}{i}, \frac{1}{k} \right\}, \max \left\{ \frac{1}{j}, \frac{1}{\ell} \right\} \right\}$$

Prove that the iterated limits $\lim_{i \rightarrow \infty} \lim_{j \rightarrow \infty} x_{ij}$, $\lim_{j \rightarrow \infty} \lim_{i \rightarrow \infty} x_{ij}$ exist and are equal.

Problem 3 (Fa02). Suppose A and M are $n \times n$ matrices over $\mathbb{C}$, A is invertible and $AMA^{-1} = M^2$. Prove the nonzero eigenvalues of M are roots of unity.

Problem 4 (Fa02, Fa09). Evaluate the integrals

$$I_n = \int_{C_n} \frac{1}{z^3 \sin z} dz \quad n = 0, 1, \dots$$

where C_n is the circle $|z| = (n + \frac{1}{2})\pi$, with the counterclockwise orientation.

Problem 5 (Fa02, Fa06). Let $\mathbb{F}$ be a field such that the additive group of $\mathbb{F}$ is finitely generated as a group. Prove that $\mathbb{F}$ is finite.

Problem 6 (Fa02). Let p be a polynomial over $\mathbb{R}$ of positive degree. Define the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $f(x, y) = (p(x + y), p(x - y))$. Prove that the derivative $Df(x, y)$ is invertible for an open dense set of points (x, y) in $\mathbb{R}^2$.

Problem 7 (Fa02). Let the $n \times n$ real matrix A be diagonalizable and have a one-dimensional null space. Prove that a nonzero left null vector of A cannot be orthogonal to a nonzero right null vector of A .

Problem 8 (Fa02). Let α be a number in $(0, \pi/2)$. Prove that the function $f(z) = e^{-1/z}$ is uniformly continuous in $S_\alpha = \{z \mid 0 < |z| \leq 1, |\arg z| \leq \alpha\}$, a sector of the complex plane.

Problem 9 (Fa02). Let G be a group of order 112. Prove that G has a non-trivial normal subgroup.

Problem 10 (Fa02). Let S be a subset of $\mathbb{R}$. Let C be the set of points x in $\mathbb{R}$ with the property that $S \cap (x - \delta, x + \delta)$ is uncountable for every $\delta > 0$. Prove that $S \setminus C$ is finite or countable.

Problem 11 (Fa02). List all groups of order ≤ 6 up to isomorphism. Find a group of order 120 that contains as a subgroup an isomorphic copy of each of them. Prove that no group of order < 120 has the preceding property.

Problem 12 (Fa02). Let k be an integer larger than 1. Find all entire functions f that satisfy

$$f(z^k) = (f(z))^k$$

Problem 13 (Fa02). Let V and W be vector spaces over a field $\mathbb{F}$. Assume $A : V \rightarrow W$ and $B : V \rightarrow W$ are linear transformations such that A has rank at least 2, and for every vector v in V the vectors Av and Bv are linearly dependent. Prove that the linear transformations A and B are linearly dependent.

Problem 14 (Fa02). Let n be a positive integer. Find a group of fractional linear transformations of $\mathbb{C}$ that fixes the two points $z = 1$ and $z = -1$ and has order n .

Problem 15 (Fa02). Find the most general continuously differentiable function $g : \mathbb{R} \rightarrow (0, \infty)$ such that the function $h(x, y) = g(x)g(y)$ on $\mathbb{R}^2$ is constant on each circle with center $(0, 0)$.

Problem 16 (Fa02). Let A and B be $n \times n$ matrices over $\mathbb{R}$ such that $A + B$ is invertible. Prove that

$$A(A + B)^{-1}B = B(A + B)^{-1}A$$

Problem 17 (Fa02, Sp14). Evaluate $\int_{-\infty}^{\infty} \frac{\cos x}{(1 + x^2)^3} dx$.

Problem 18 (Fa02). Let $\mathbb{K}$ be a field and $\mathbb{L} \subset \mathbb{K}$ a subfield containing $\{a^2 \mid a \in \mathbb{K}\}$. Prove that if the characteristic of $\mathbb{K}$ is not 2 then $\mathbb{L} = \mathbb{K}$ and that the same conclusion holds for finite fields $\mathbb{K}$ of characteristic 2, but not in general for fields of characteristic 2.